

"99.9% of science is boring."
- Makise Kurisu

THEORETICAL COMPUTER SCIENCE

August 22, 2026

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CONTENTS

1	Mathematical Notions and Terminology	1
1.1	Sets	1

1 MATHEMATICAL NOTIONS AND TERMINOLOGY

1.1 Sets

Definition 1.1 (Sets). A *set* is a group of objects represented as a unit. Sets may contain the type of the object, including numbers, symbols and even other sets. The objects in a set are called its *elements* or *members*. Sets may be described formally in several ways. Such as

$$S = \{7, 21, 57\}$$

The symbols $\in$ and $\notin$ denote set membership and non membership. We write $7 \in \{7, 21, 57\}$ and $8 \notin \{7, 21, 57\}$.